

1.

$$m(x) = \frac{1}{N} \sum_{i=1}^N x_i$$

$$m(a+bX) = \frac{1}{N} \sum_{i=1}^N a + bx_i$$

since a is not indexed, $m(a+bX) = a + \frac{1}{N} \sum_{i=1}^N bx_i$

$$m(a+bX) = a + b \frac{1}{N} \sum_{i=1}^N x_i$$

$$a + b \frac{1}{N} \sum_{i=1}^N x_i = a + b m(x)$$

$$\therefore m(a+bX) = a + b m(x)$$

2.

$$\text{cov}(X, Y) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(Y))$$

$$\text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(a + by_i - m(a+bY))$$

$$m(a+bY) = a + b m(Y)$$

$$\therefore \text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(a + by_i - [a + b m(Y)])$$

$$\text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(a + by_i - a - b m(Y))$$

$$\text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(by_i - b m(Y))$$

$$\text{cov}(X, a+bY) = \frac{1}{N} \sum_{i=1}^N b(x_i - m(X))(y_i - m(Y))$$

$$\text{cov}(X, a+bY) = b \frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(Y))$$

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(X))(y_i - m(Y)) = \text{cov}(X, Y)$$

$$\therefore \text{cov}(X, a+bY) = b \text{cov}(X, Y)$$

3.

$$\text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X) = b^2 s^2$$

$$\text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N (a + bx_i - m(a+bX))(a + bx_i - m(a+bX))$$

$$\text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N (a + bx_i - m(a+bX))^2$$

$$\text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N (a + bx_i - [a + b m(X)])^2$$

$$\text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N (a + bx_i - a - b m(X))^2$$

$$\text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N (bx_i - b m(X))^2$$

$$\text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N (bx_i - b m(X))^2$$

$$\text{cov}(a+bX, a+bX) = \frac{1}{N} \sum_{i=1}^N b^2 (x_i - m(X))^2$$

$$\text{cov}(a+bX, a+bX) = b^2 \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2$$

$$\frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2 = \text{cov}(X, X)$$

$$\therefore \text{cov}(a+bX, a+bX) = b^2 \text{cov}(X, X)$$

$$s^2 = \frac{1}{N} \sum_{i=1}^N (x_i - m(X))^2 \therefore \text{cov}(X, X) = s^2$$

$$\therefore \text{cov}(a+bX, a+bX) = b^2 s^2$$

4. $m(g(X)) = g(m(X))$

$$g(X) = x^2$$

$$m(g(X)) = \frac{1}{N} \sum_{i=1}^N x_i^2$$

$$g(m(X)) = \left(\frac{1}{N} \sum_{i=1}^N x_i \right)^2 \neq \frac{1}{N} \sum_{i=1}^N x_i^2$$

It is not always true